



Xi'an Jiaotong-Liverpool University

西交利物浦大學

INT305 Machine Learning

Lecture 5

Neural Network and Back Propagation

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Neural network

Neural Network: without the brain stuff 不涉及大脑相关内容

(Before) Linear score function: $f = Wx$

Neural network



Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$

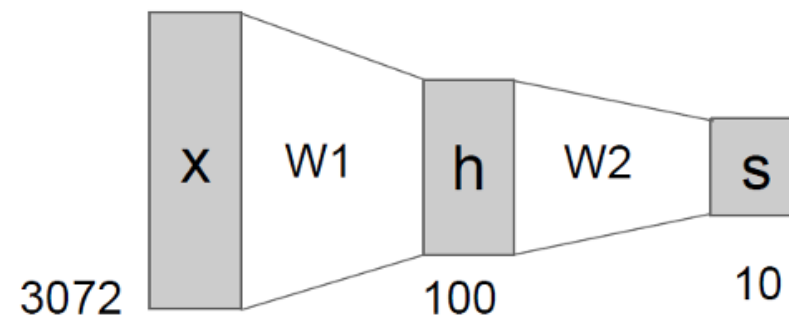


Neural network

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$





Neural network

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$

or 3-layer Neural Network:

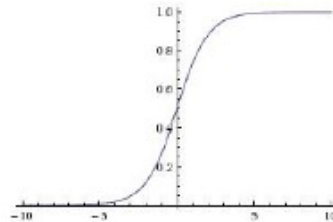
$$f = W_3 \max(0, W_2 \max(0, W_1 x))$$

Activation functions

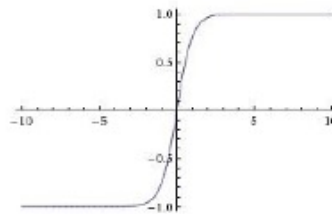
Activation Functions

Sigmoid

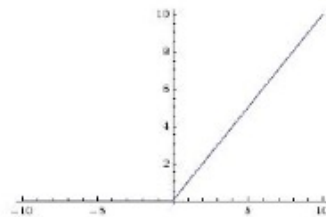
$$\sigma(x) = 1/(1 + e^{-x})$$



tanh $\tanh(x)$

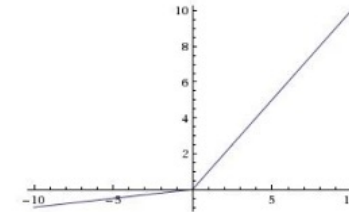


ReLU $\max(0, x)$



Leaky ReLU

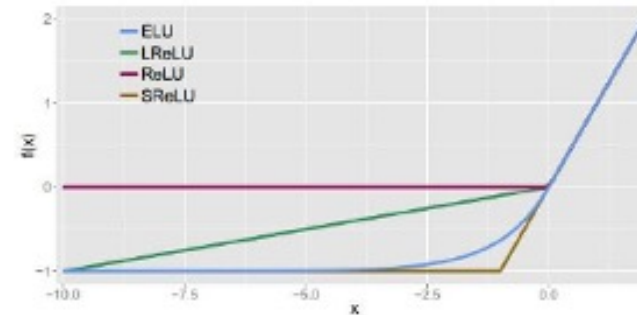
$$\max(0.1x, x)$$



Maxout $\max(w_1^T x + b_1, w_2^T x + b_2)$

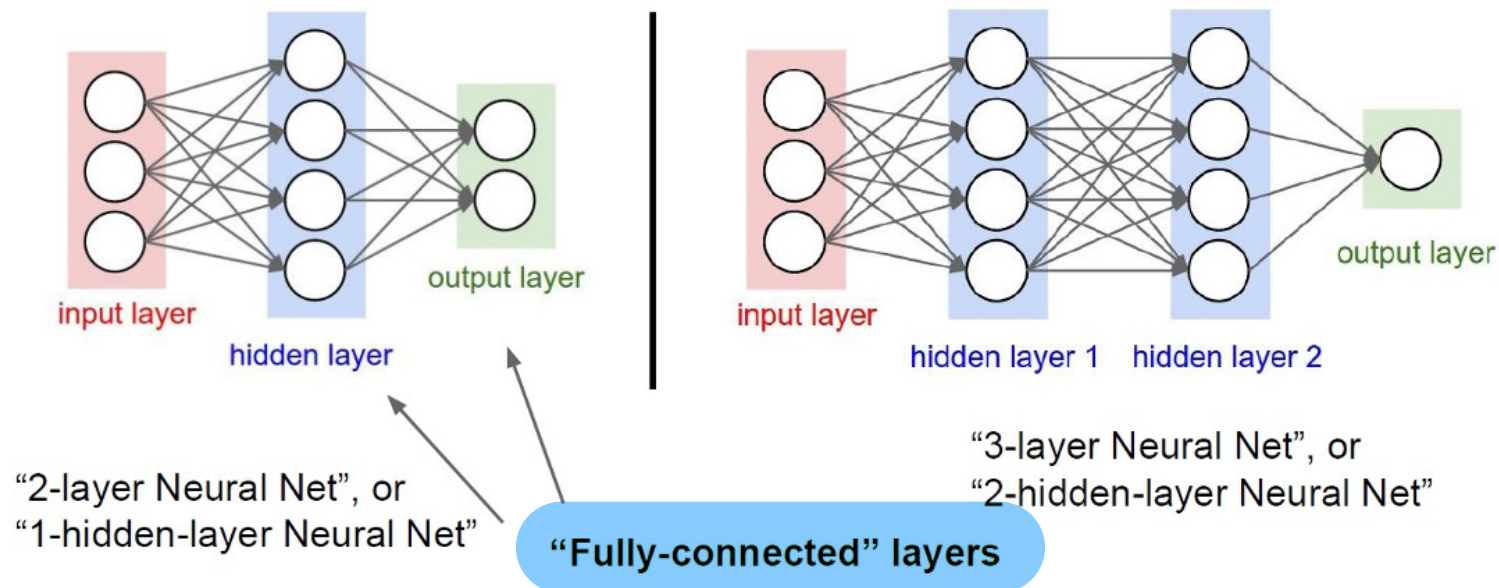
ELU

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(\exp(x) - 1) & \text{if } x \leq 0 \end{cases}$$



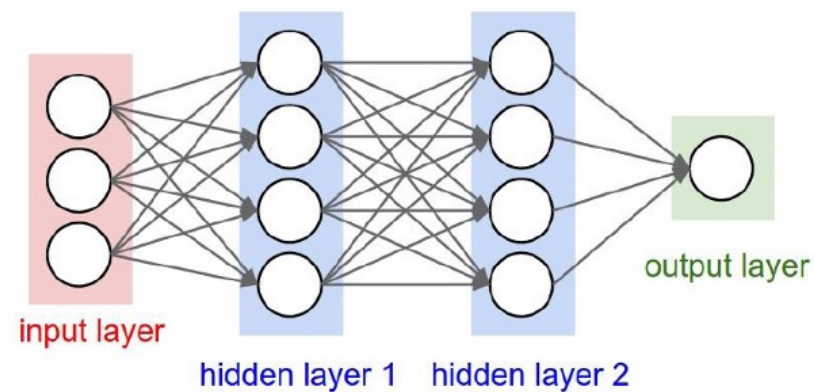
Neural network

Neural Networks: Architectures



Neural network

Example Feed-forward computation of a Neural Network



```
# forward-pass of a 3-layer neural network:  
f = lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)  
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)  
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)  
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)  
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

Gradient Descent

Where we are...

$$s = f(x; W) = Wx$$

scores function

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

SVM loss

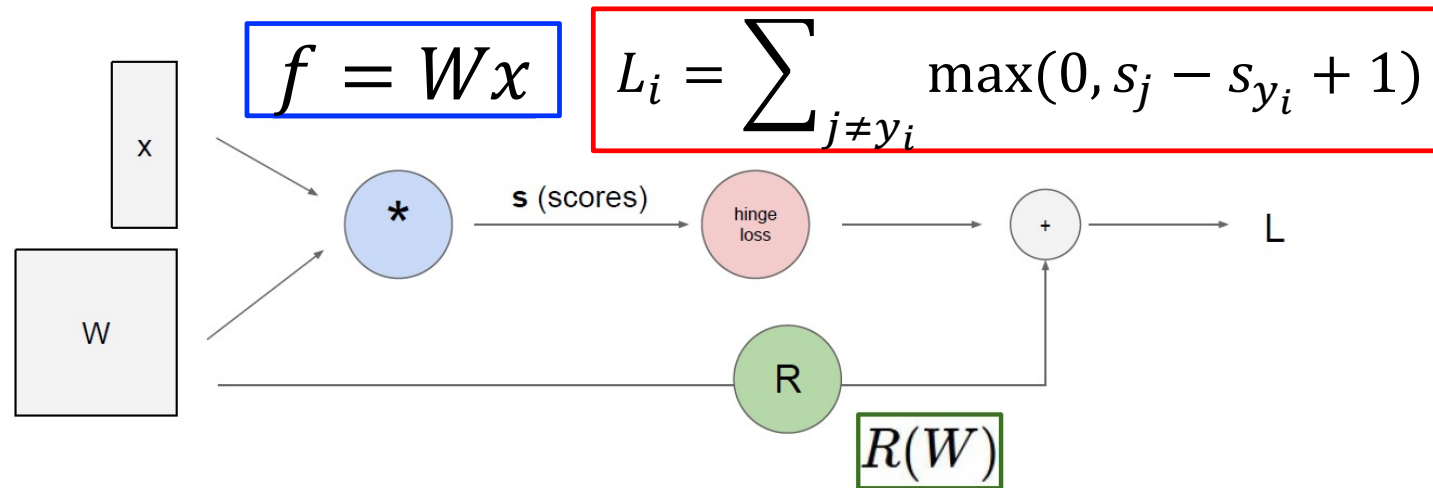
$$L = \frac{1}{N} \sum_{i=1}^N L_i + \sum_k W_k^2$$

data loss + regularization

want $\nabla_W L$

Computational Graph

Computational Graph



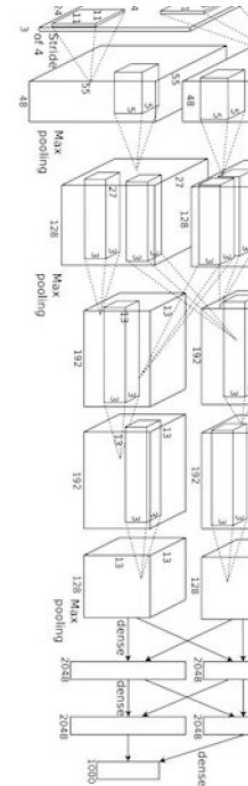
Computational Graph

Convolutional Network
(AlexNet)

input image

weights

loss

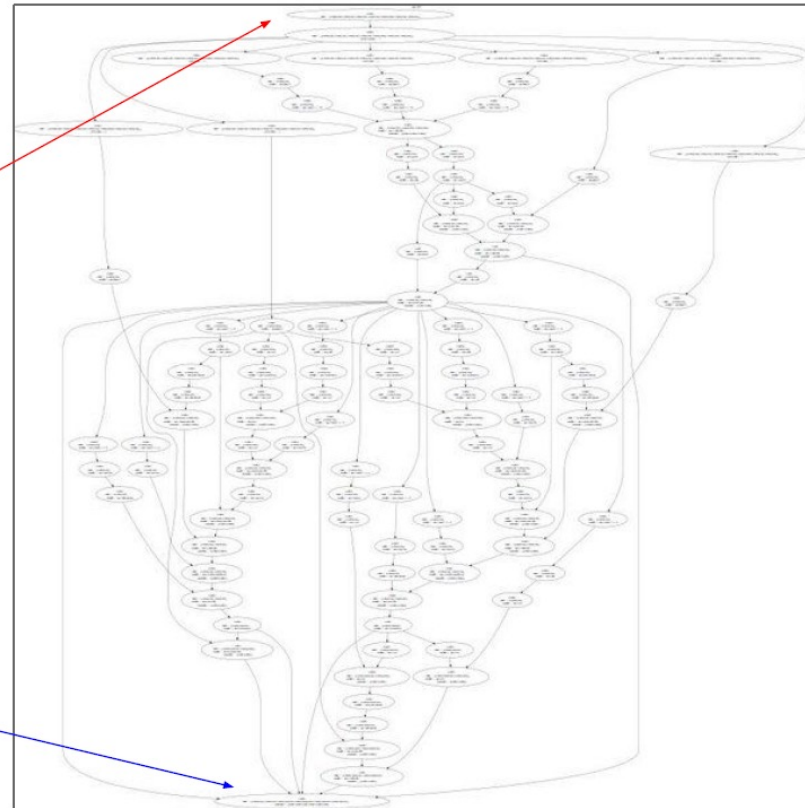


Computational Graph

Neural Turing Machine

input tape

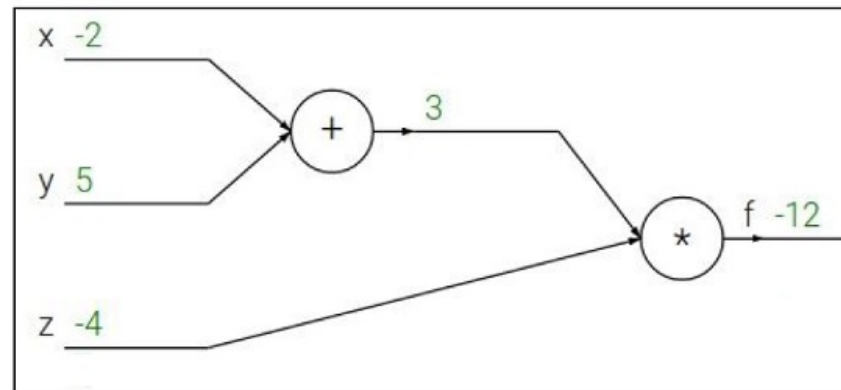
loss



Example 1

$$f(x, y, z) = (x + y)z$$

e.g. $x = -2, y = 5, z = -4$



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$$q = x + y \quad \frac{\partial q}{\partial x} = 1, \frac{\partial q}{\partial y} = 1$$

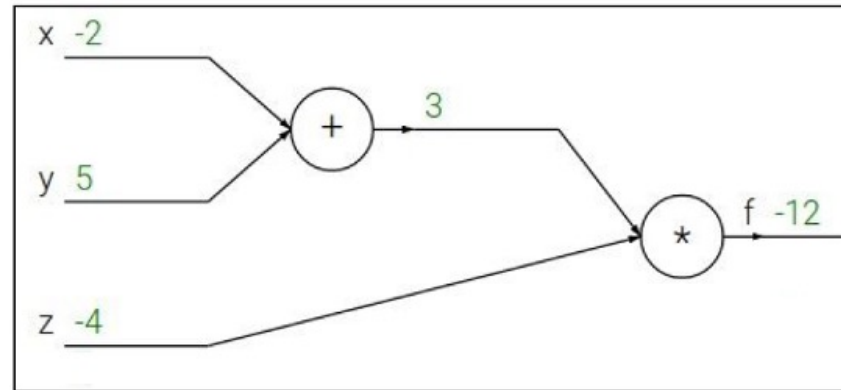
$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = z \cdot 1 = -4$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = z \cdot 1 = -4$$

$$\frac{\partial f}{\partial z} = q = x + y = 3$$



Example 1

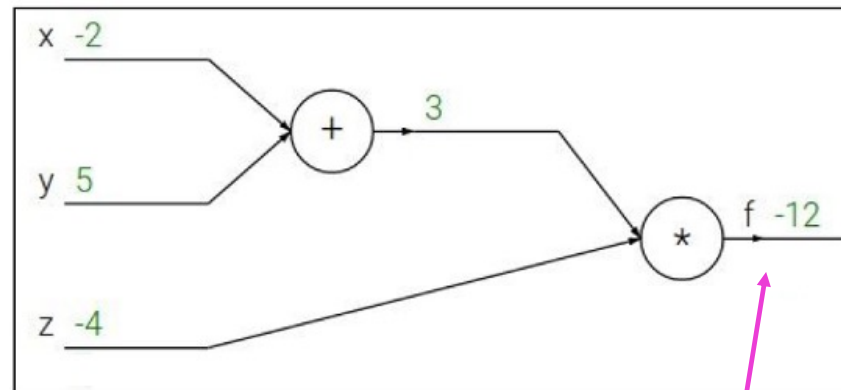
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$$\frac{\partial f}{\partial f}$$

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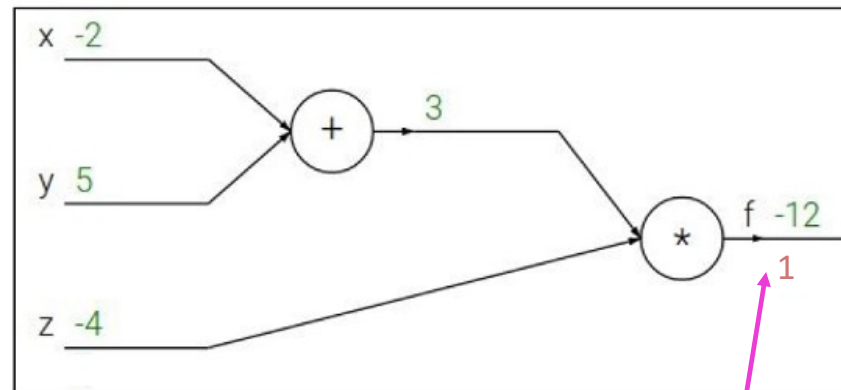
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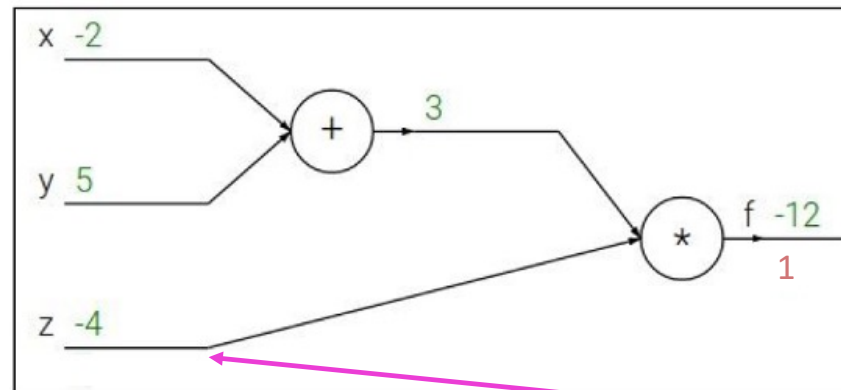
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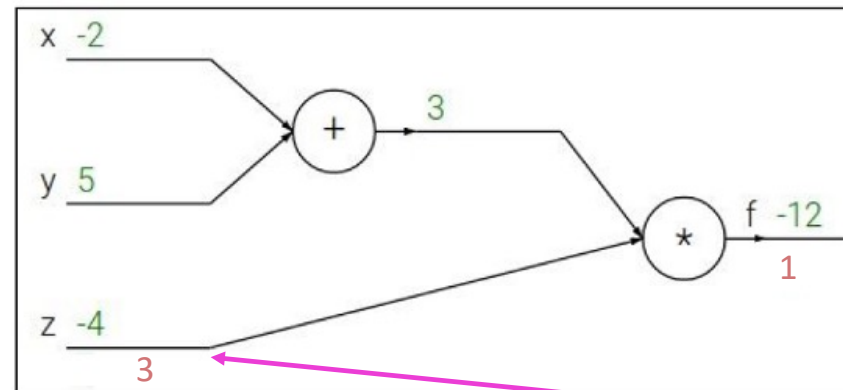
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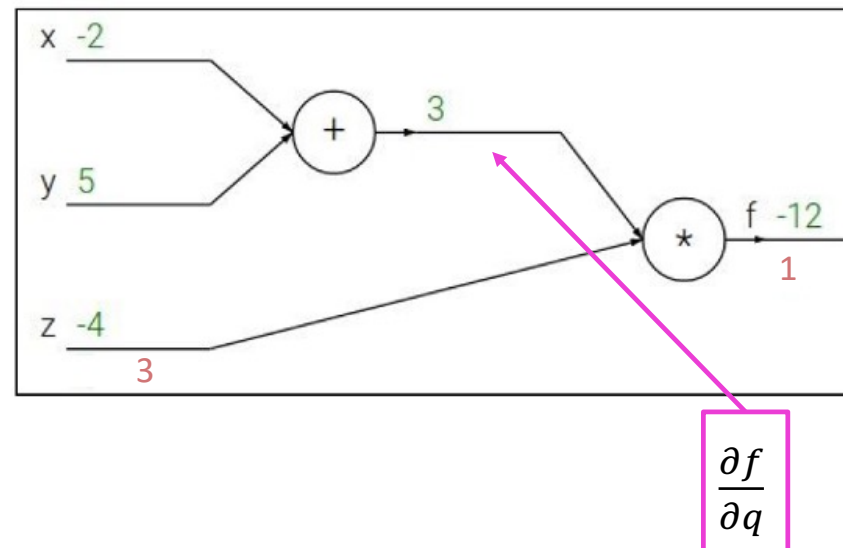
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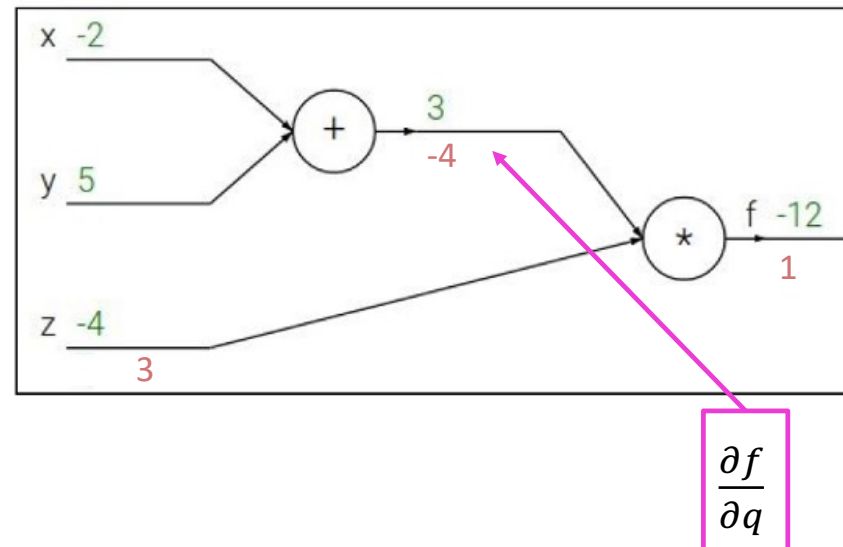
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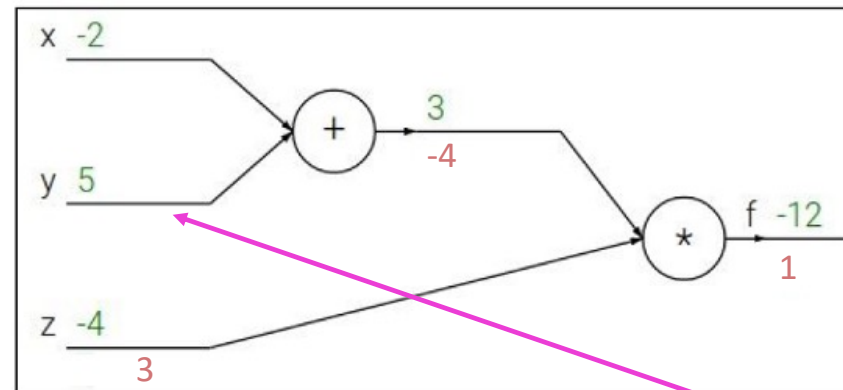
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$$\frac{\partial f}{\partial y}$$

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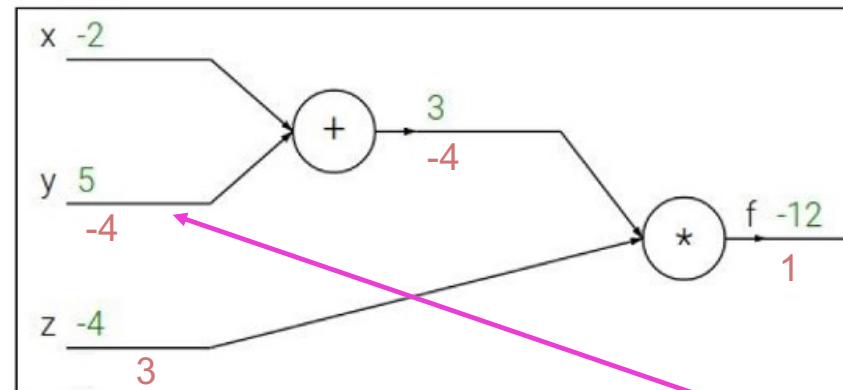
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Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



$$\frac{\partial f}{\partial y}$$

Chain rule:

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$$

Example 1

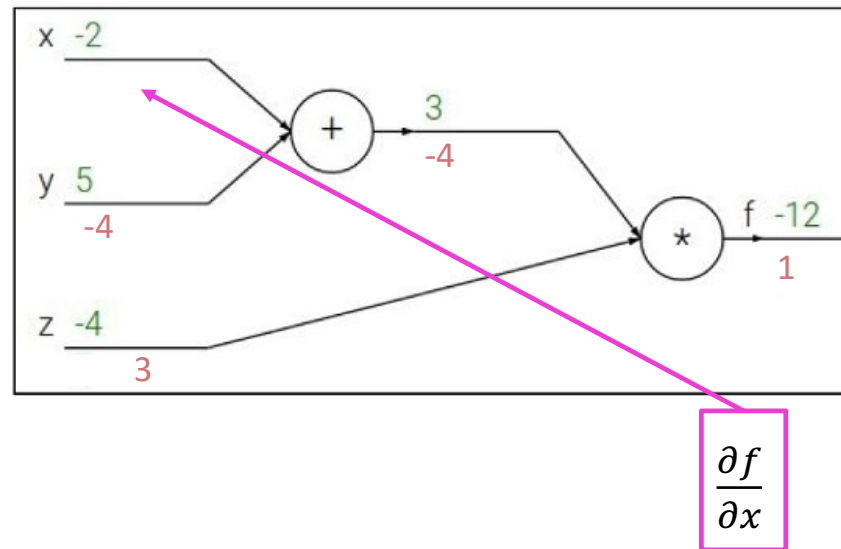
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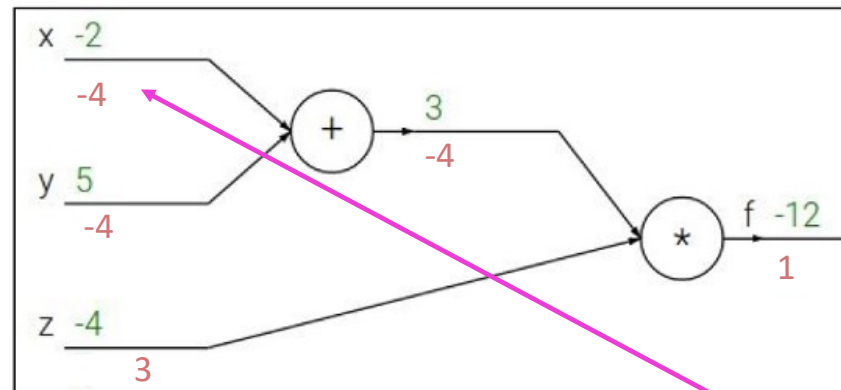
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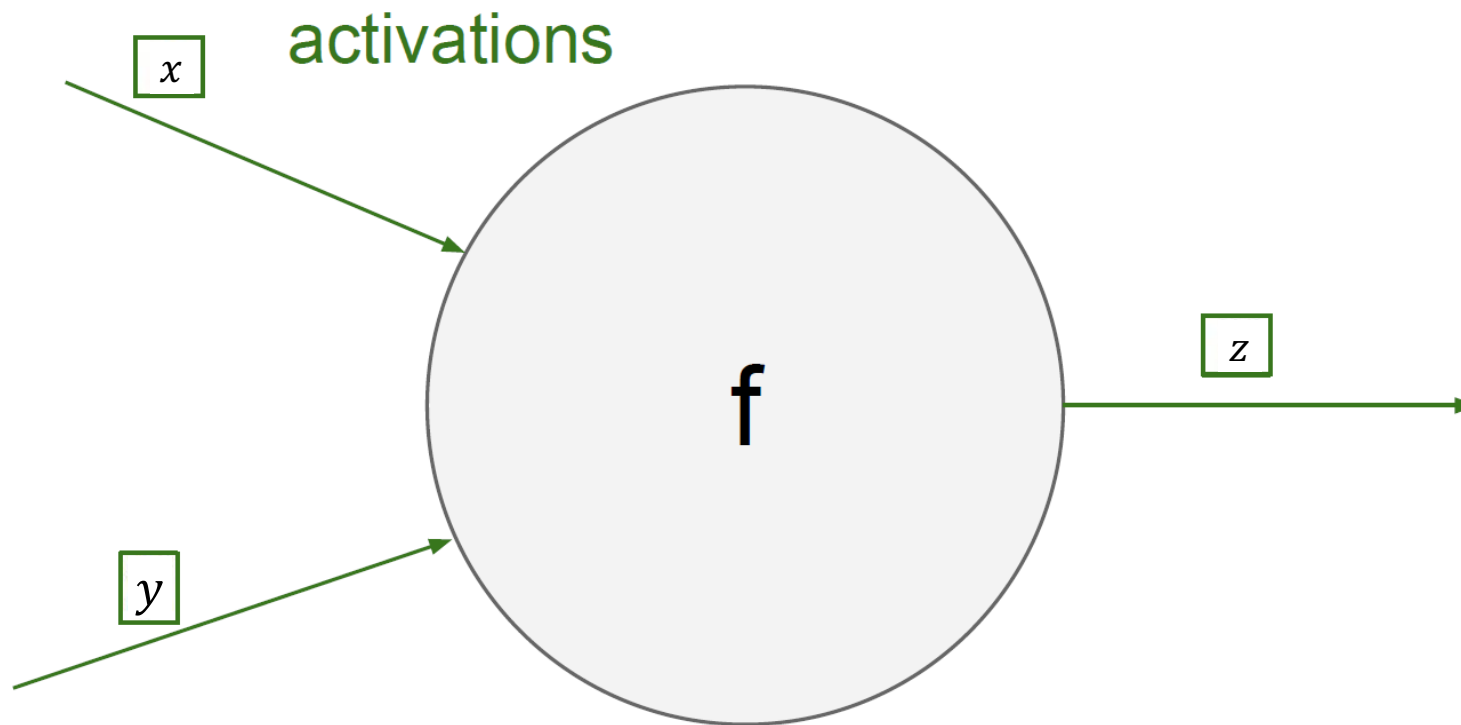
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Chain rule:

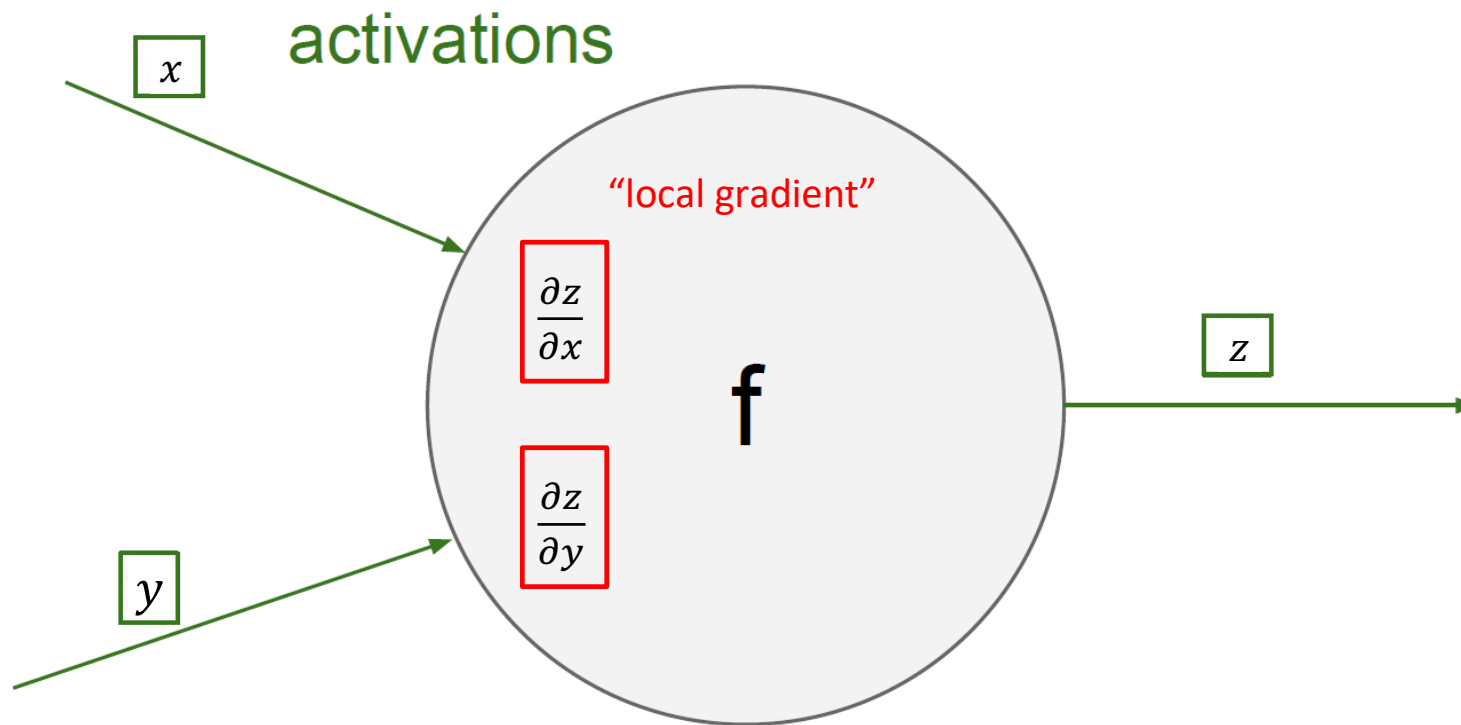
$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$$



Chain rule

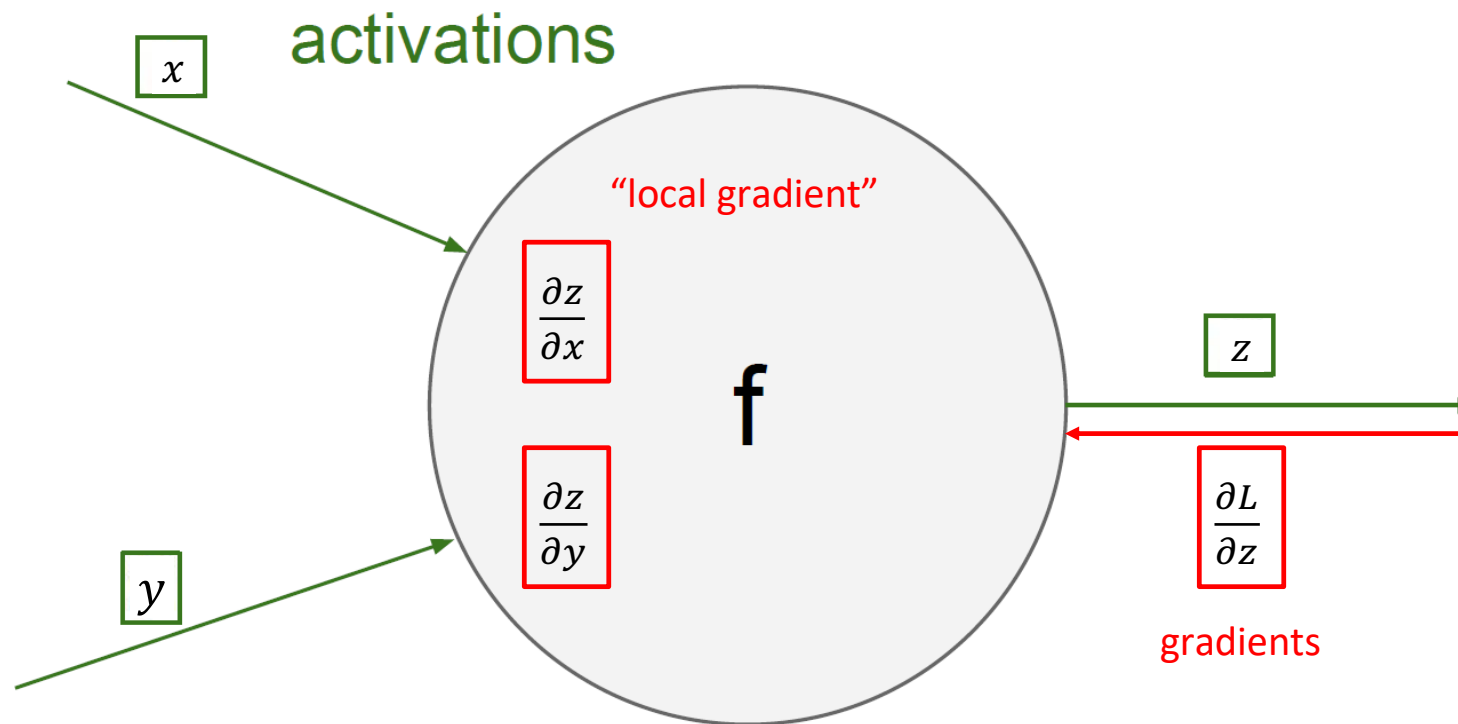


Chain rule

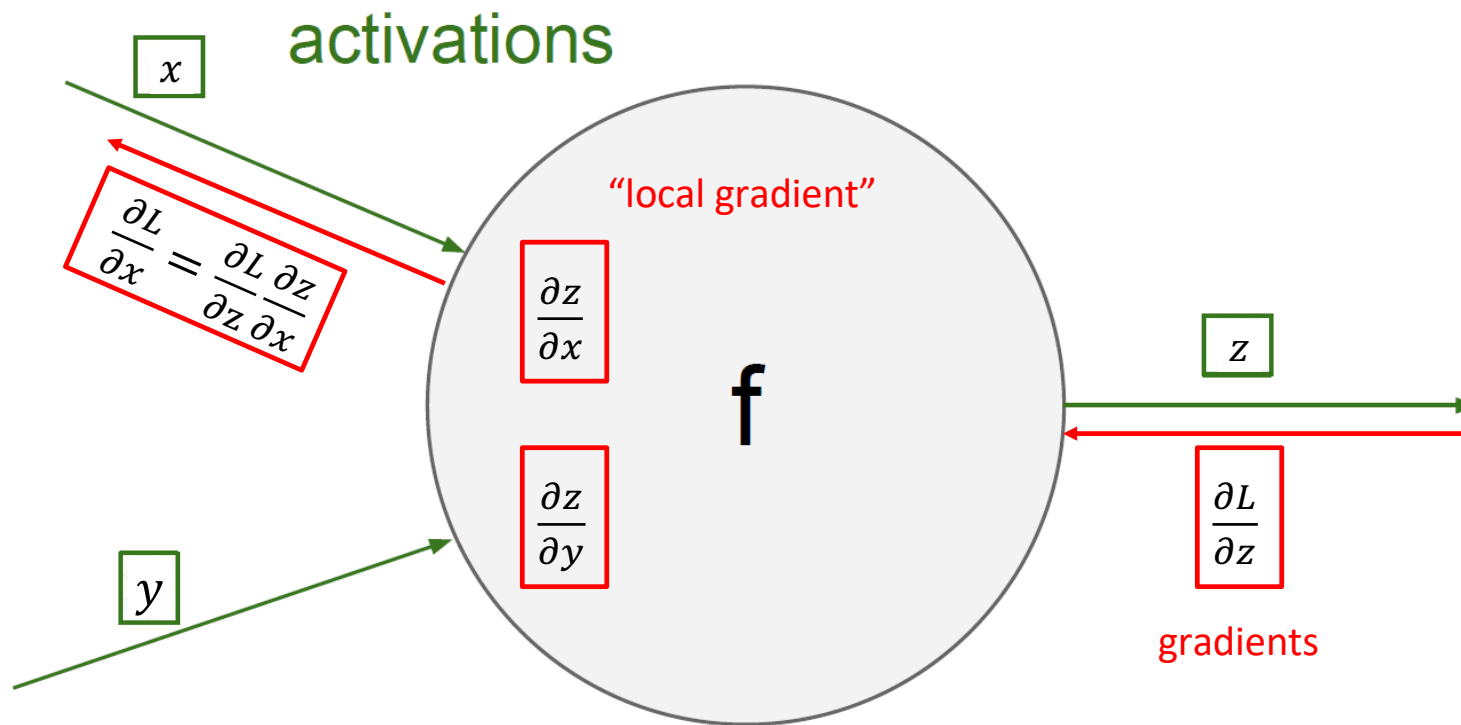




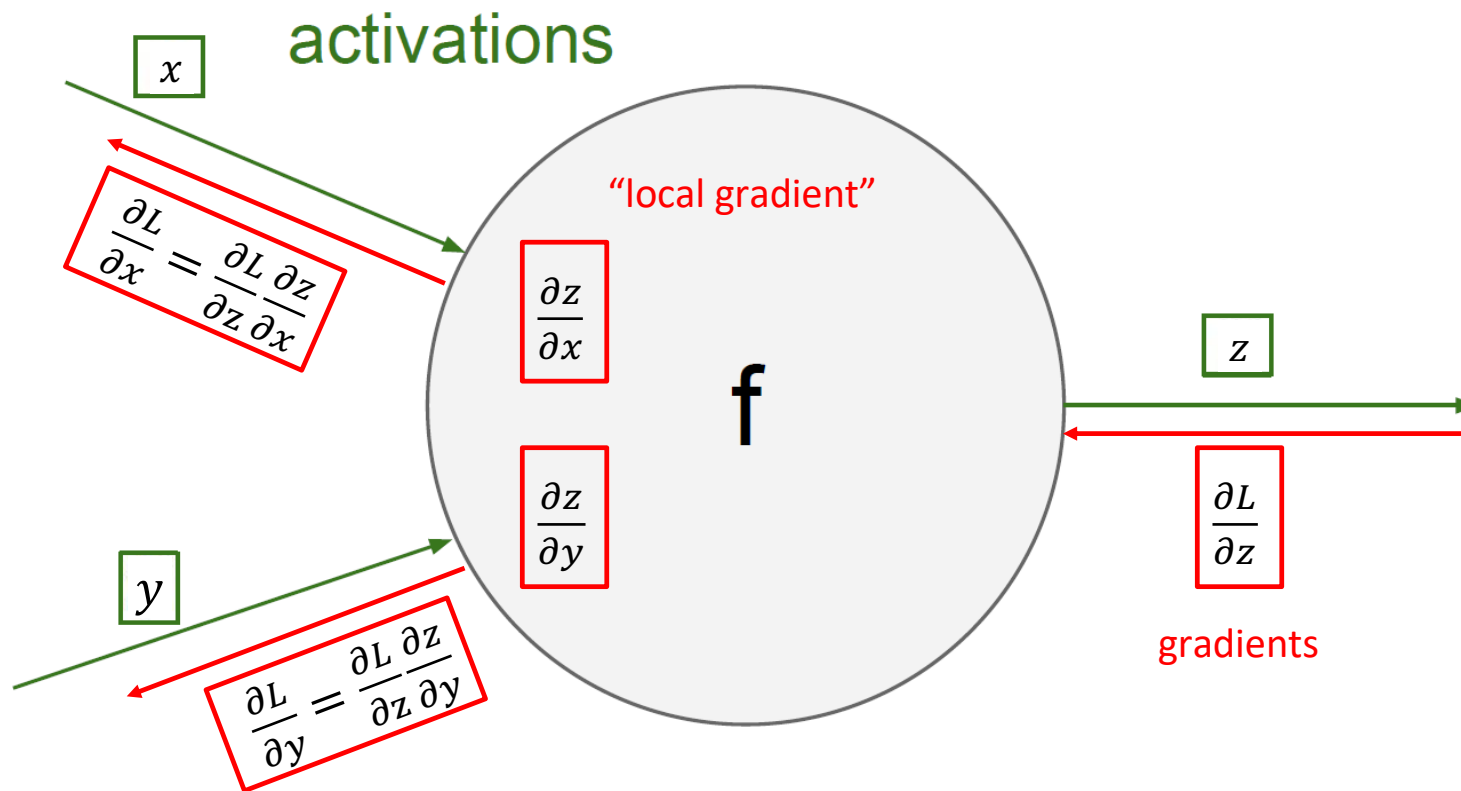
Chain rule



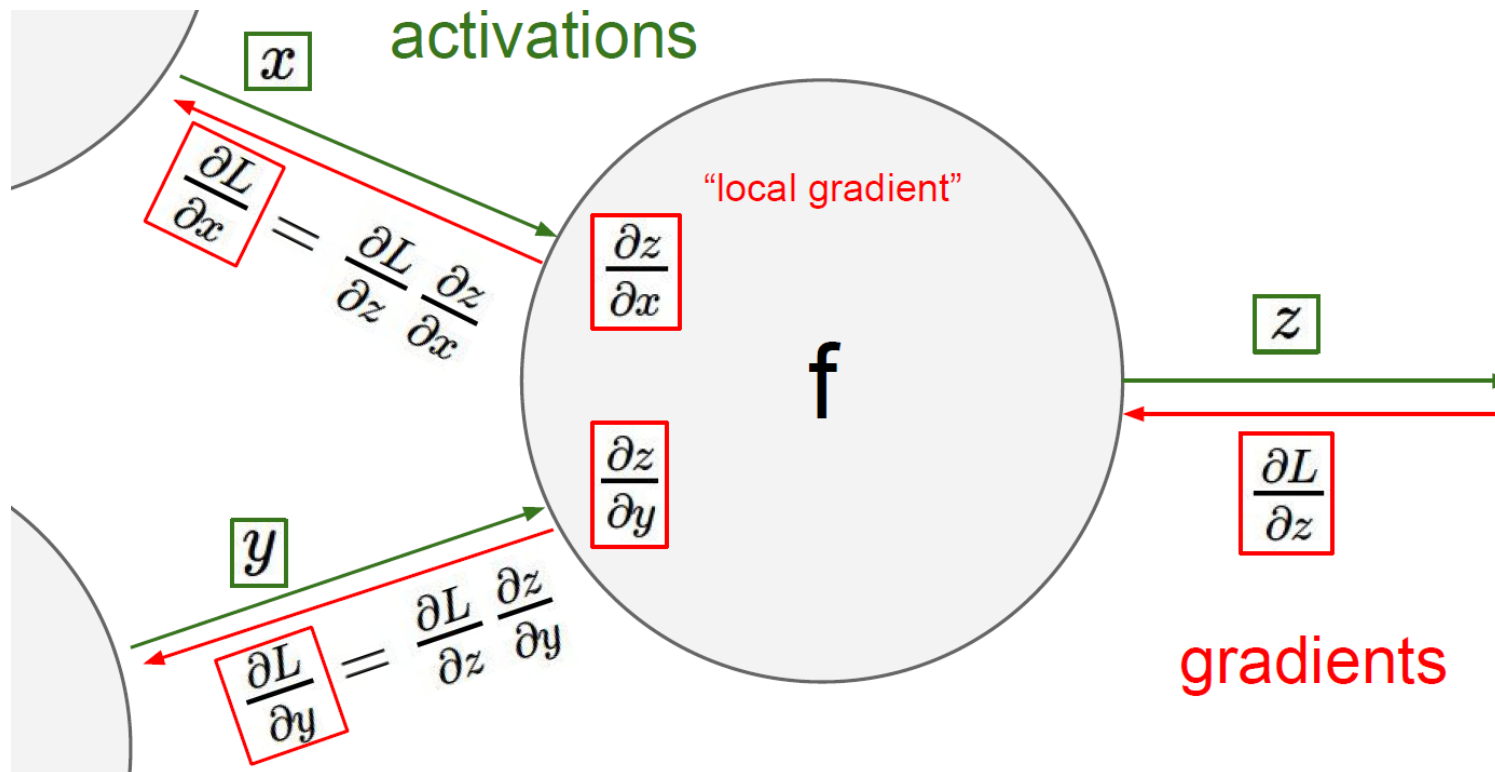
Chain rule



Chain rule

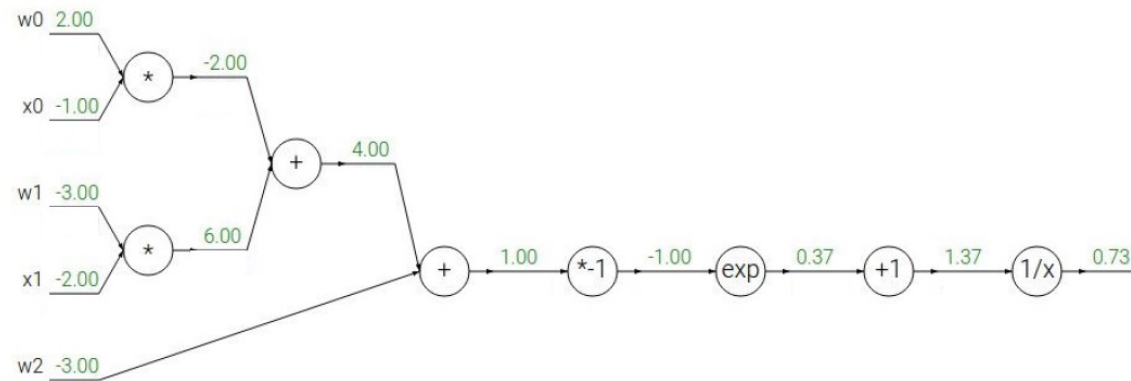


Chain rule



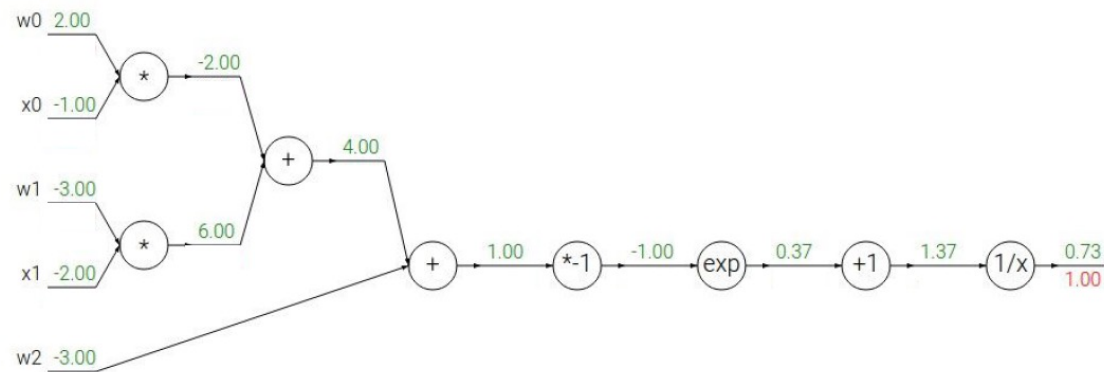
Chain rule

Another example:
$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$$



Chain rule

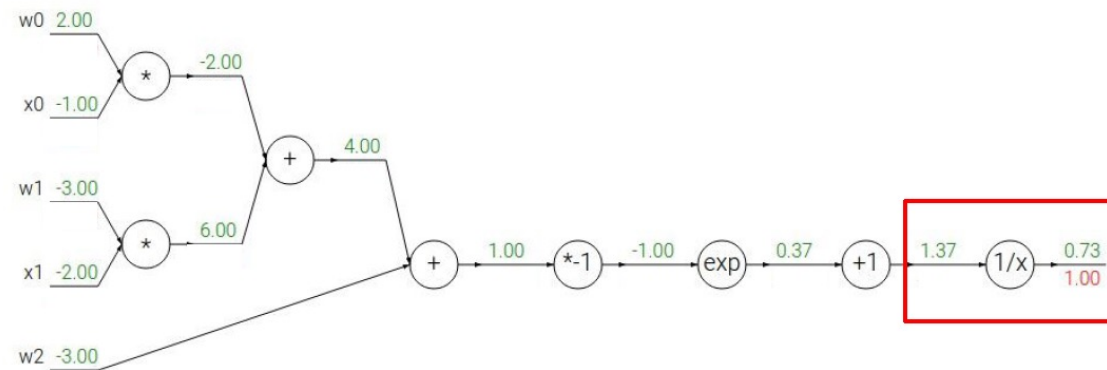
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$



$f(x) = e^x$	→	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	→	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	→	$\frac{df}{dx} = a$		$f_c(x) = c + x$	→	$\frac{df}{dx} = 1$

Chain rule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$

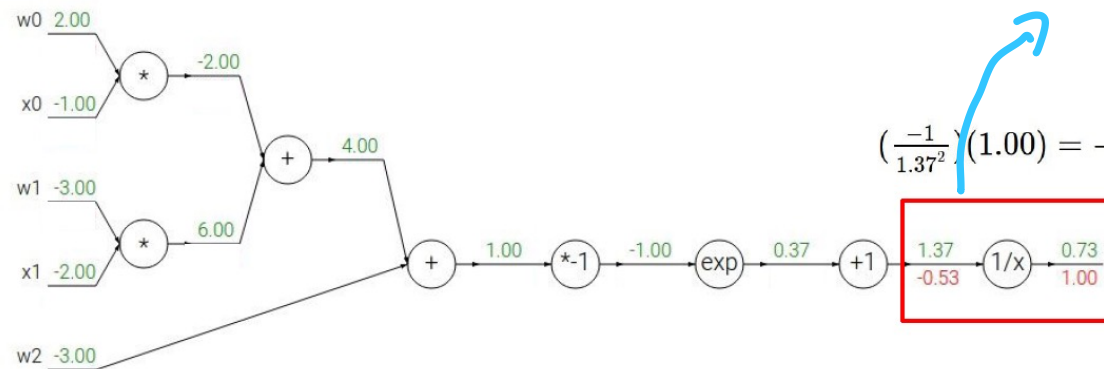


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Chain rule

Another example:

$$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2 x_2)}}$$



$$f(m) = \frac{1}{m}, m = 1.37$$

$$\frac{df}{dm} = -\frac{1}{m^2} = -0.53$$

$$\left(\frac{-1}{1.37^2}\right)(1.00) = -0.53$$

$$f(x) = e^x$$

→

$$\frac{df}{dx} = e^x$$

$$f_a(x) = ax$$

→

$$\frac{df}{dx} = a$$

$$f(x) = \frac{1}{x}$$

→

$$\frac{df}{dx} = -1/x^2$$

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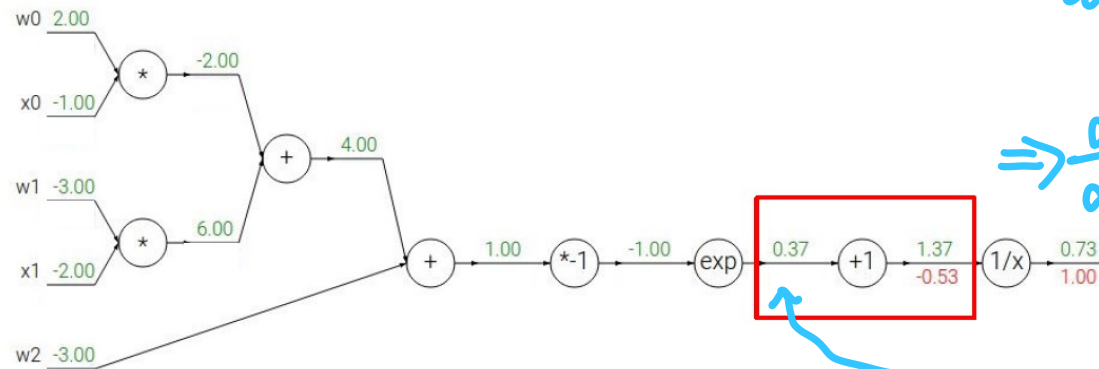
→

$$\frac{df}{dx} = 1$$

Chain rule

Another example:

$$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$



$$\frac{dm}{dq} = 1, \quad \frac{df}{dq} ?$$

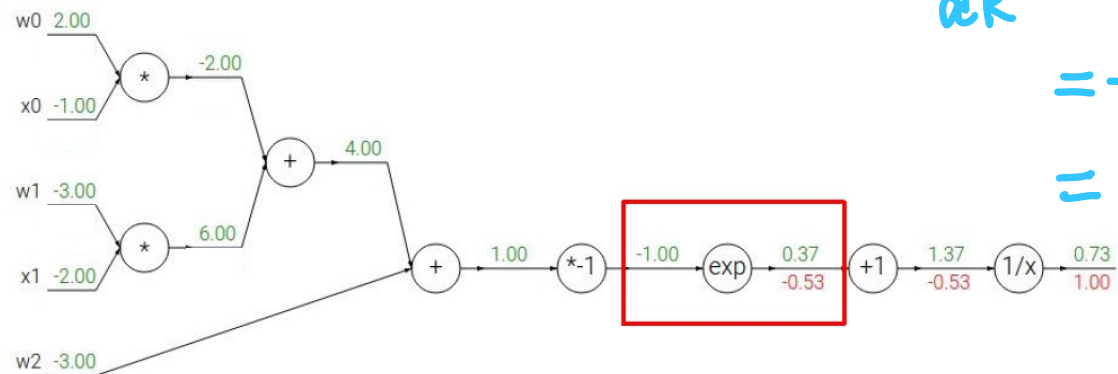
$$\Rightarrow \frac{df}{dm} \times \frac{dm}{dq} = -0.53$$

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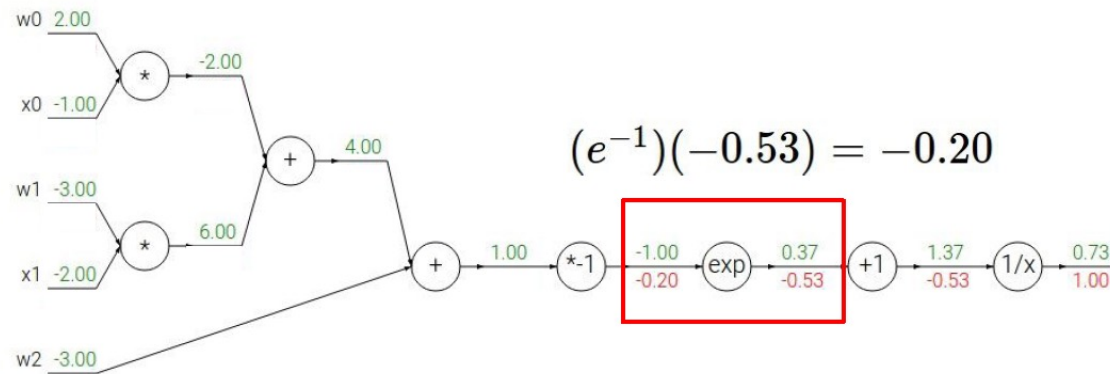
$$\begin{aligned} \frac{df}{dk} &= \frac{df}{dp} \times \frac{dp}{dk} \\ &= -0.53 \times e^k \\ &= -0.53 \times 0.37 \approx -0.2 \end{aligned}$$

$$\begin{array}{ll} f(x) = e^x & \rightarrow \frac{df}{dx} = e^x \\ f_a(x) = ax & \rightarrow \frac{df}{dx} = a \end{array}$$

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Chain rule

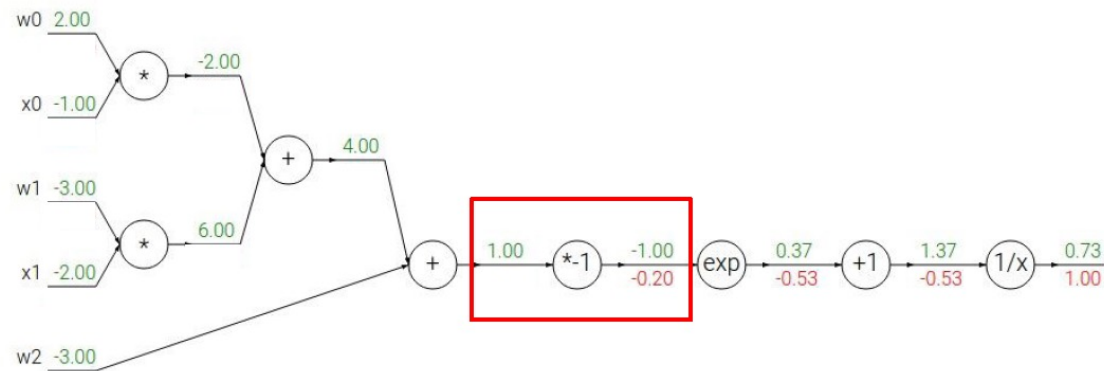
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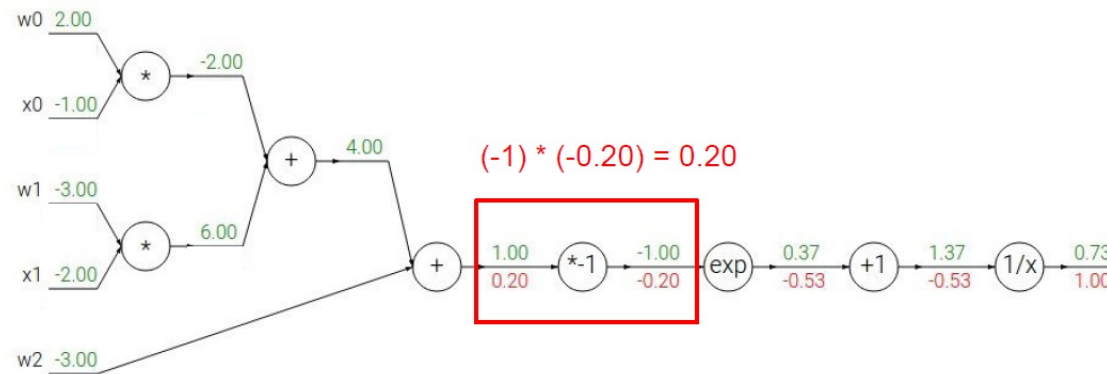


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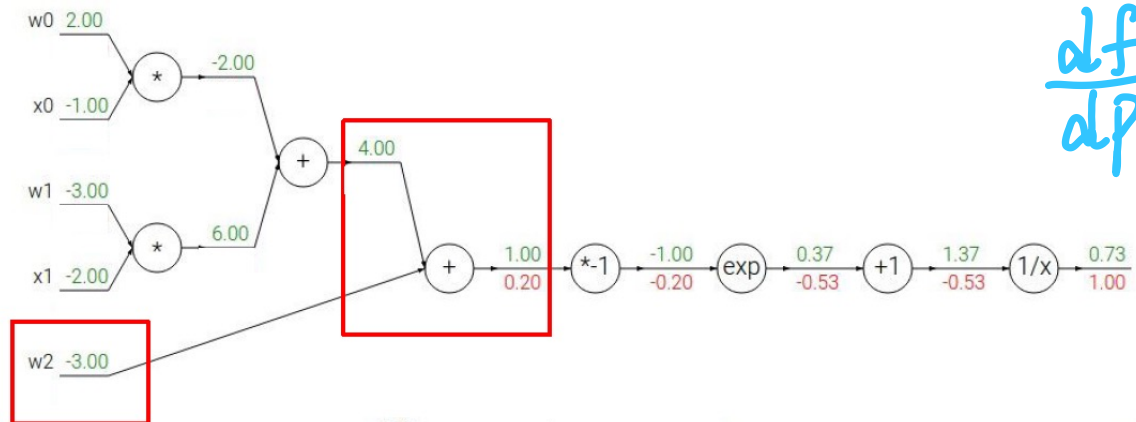
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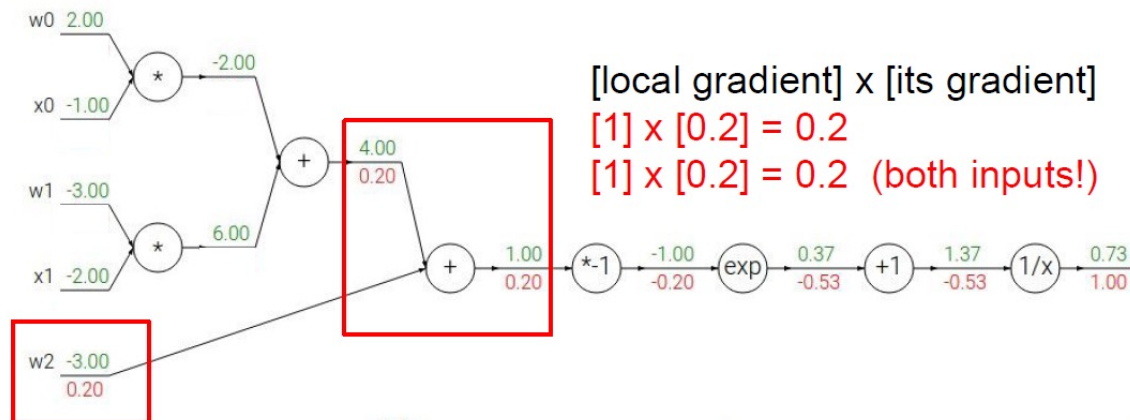


$$\frac{df}{dp} = \frac{df}{dz} \times \frac{dz}{dp}$$

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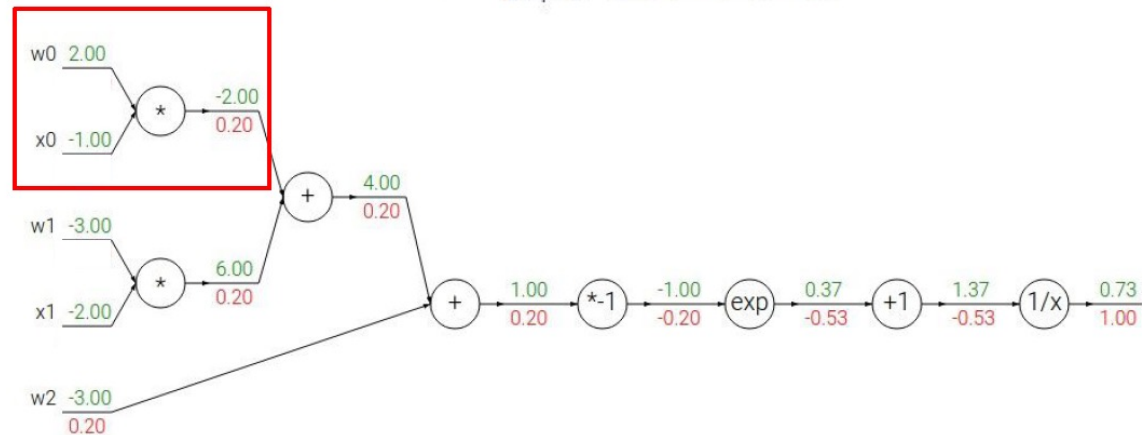


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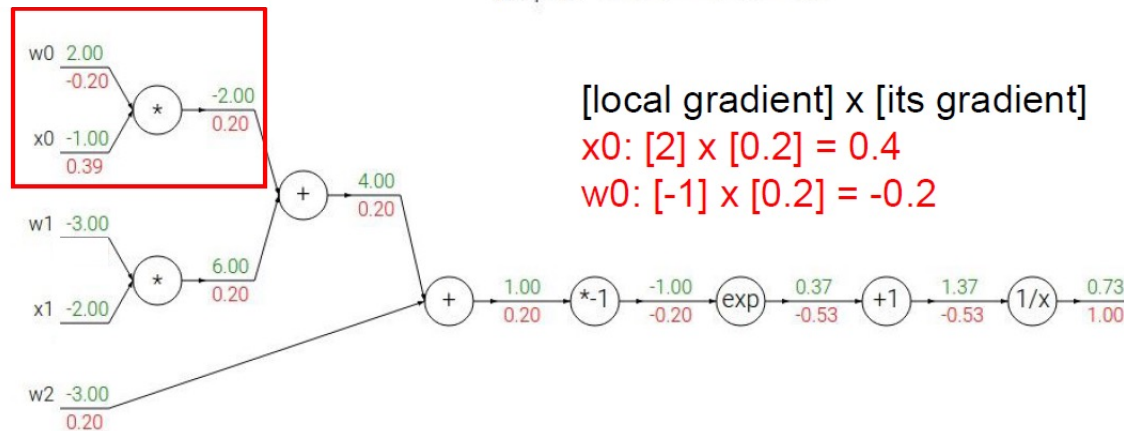


$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chain rule

Another example:

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$$



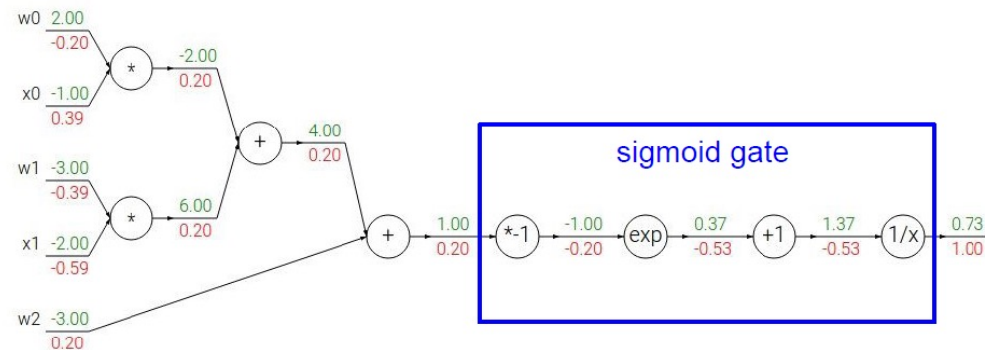
$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Sigmoid

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad \text{sigmoid function}$$

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}} \right) \left(\frac{1}{1 + e^{-x}} \right) = (1 - \sigma(x))\sigma(x)$$

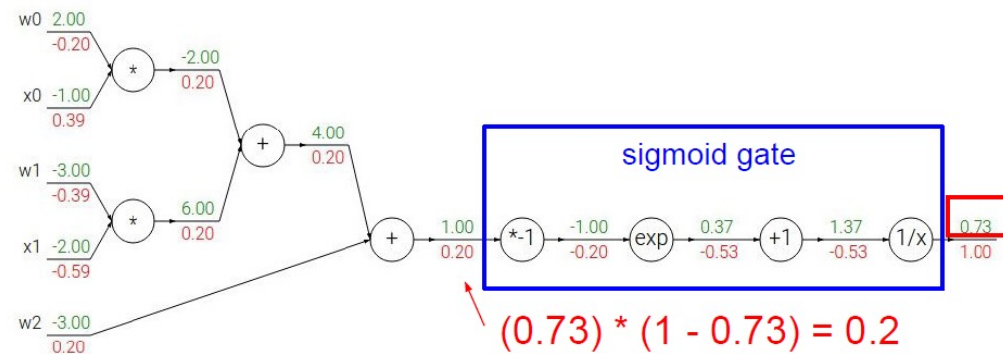


Sigmoid

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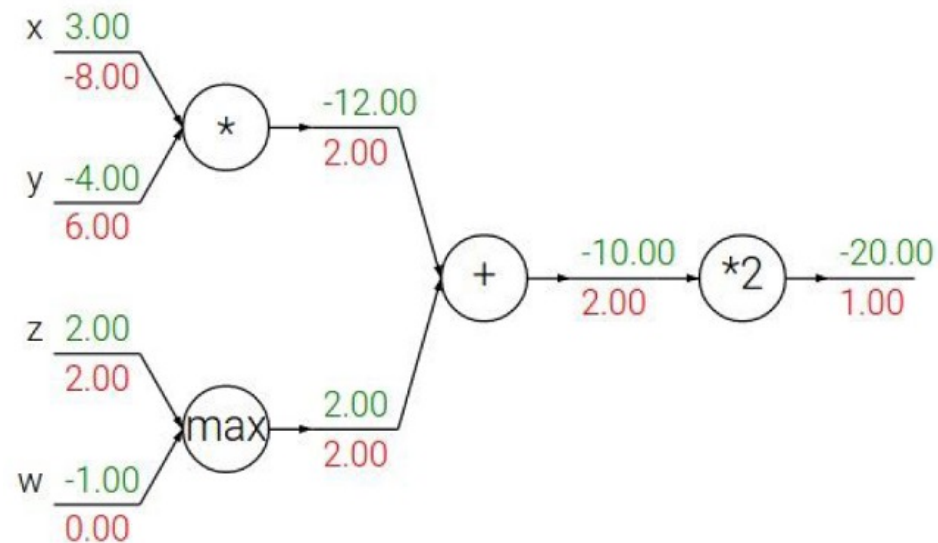
Pattern in backward flow

Patterns in backward flow

add gate: gradient distributor

max gate: gradient router

mul gate: gradient... “switcher”?



Exercise 1

Pooling units take n values x_i , $i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

1. The Lp-pooling module takes positive inputs and computes

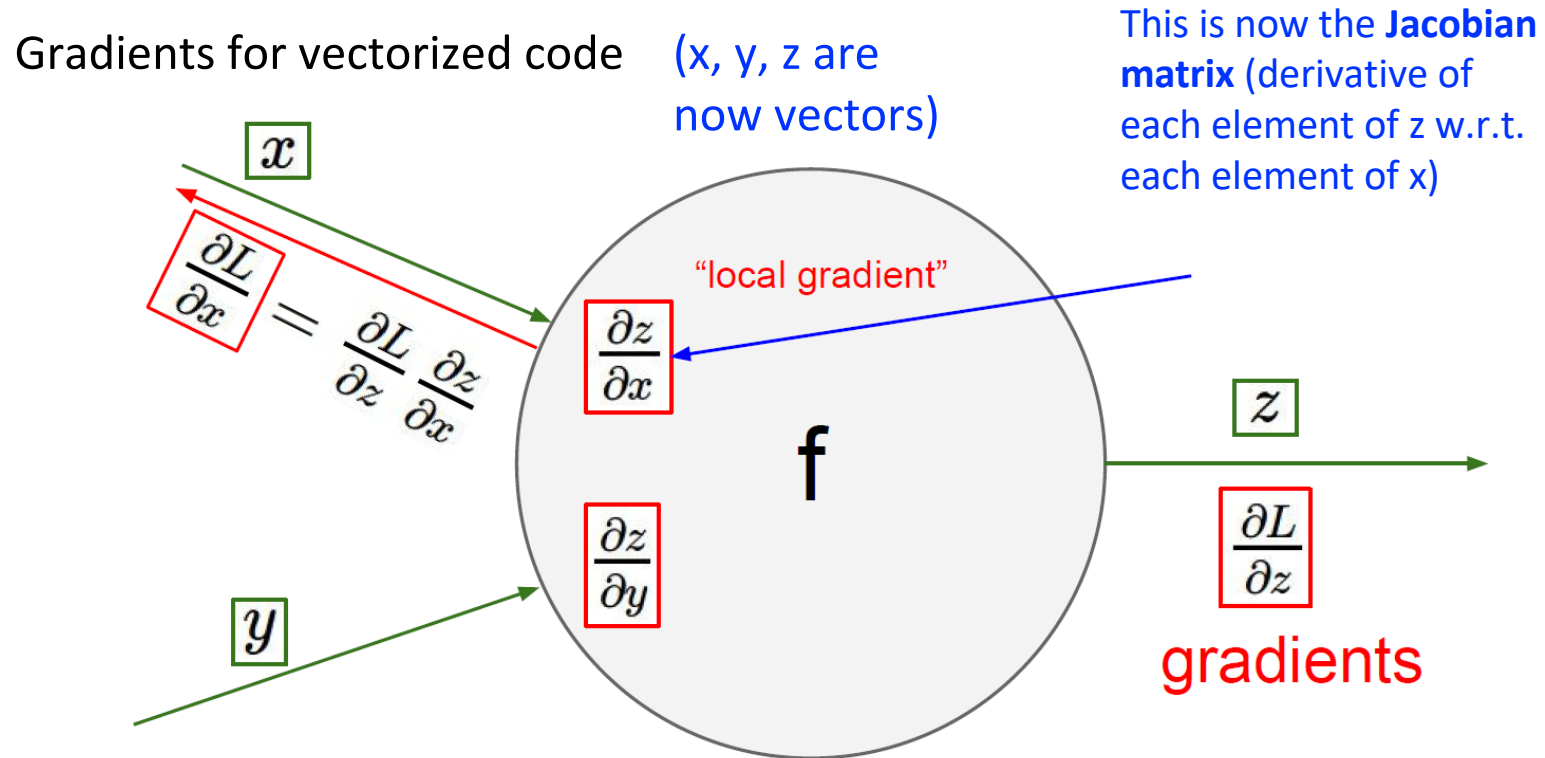
$y = (\sum_i x_i^p)^{\frac{1}{p}}$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x'_i = \frac{\partial L}{\partial x_i}$?

$$\frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y} \times \frac{\partial y}{\partial x_i} = y' \cdot \frac{\partial (\sum_i x_i^p)^{\frac{1}{p}}}{\partial x_i} = y' \cdot \frac{1}{p} (\sum_i x_i^p)^{\frac{1}{p}-1} \cdot p x_i^{p-1} = y' \cdot \frac{x_i^{p-1}}{(\sum_i x_i^p)^{\frac{p-1}{p}}}$$

2. The log-average module computes $y = \frac{1}{\beta} \ln(\frac{1}{n} \sum_i \exp(\beta x_i))$,

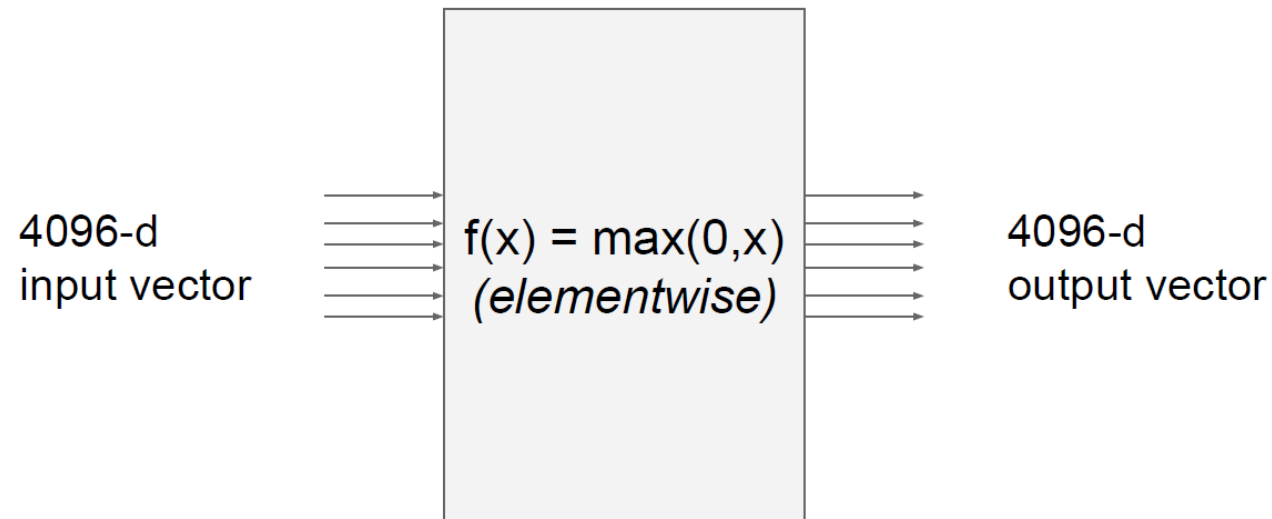
assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x'_i = \frac{\partial L}{\partial x_i}$?

Gradients for vector



Gradients for vector

Vectorized operation

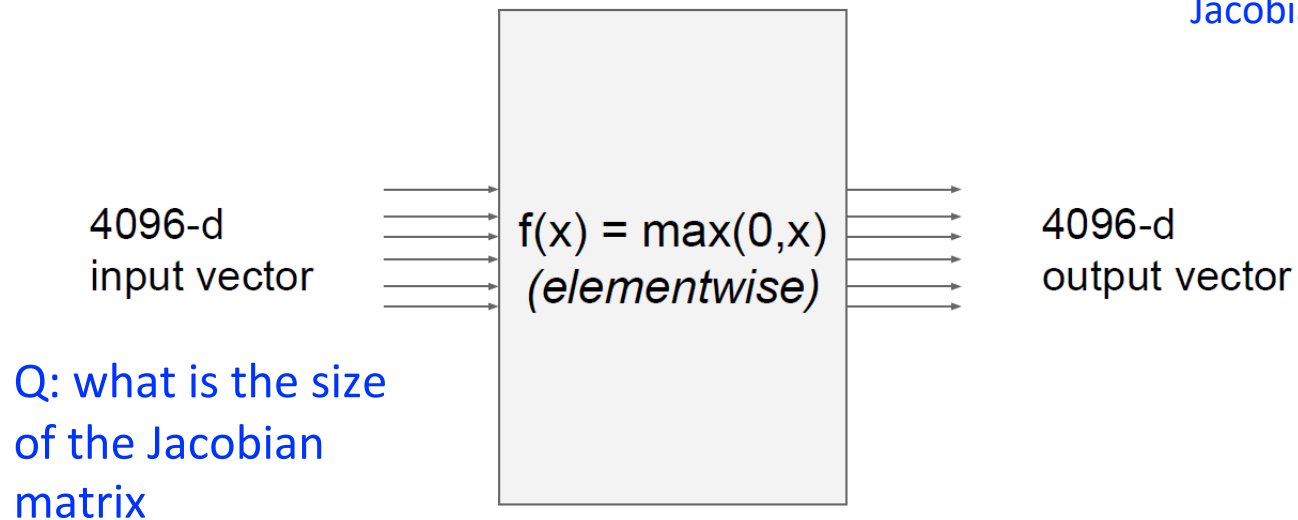


Gradients for vector

Vectorized operation

$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

Jacobian matrix

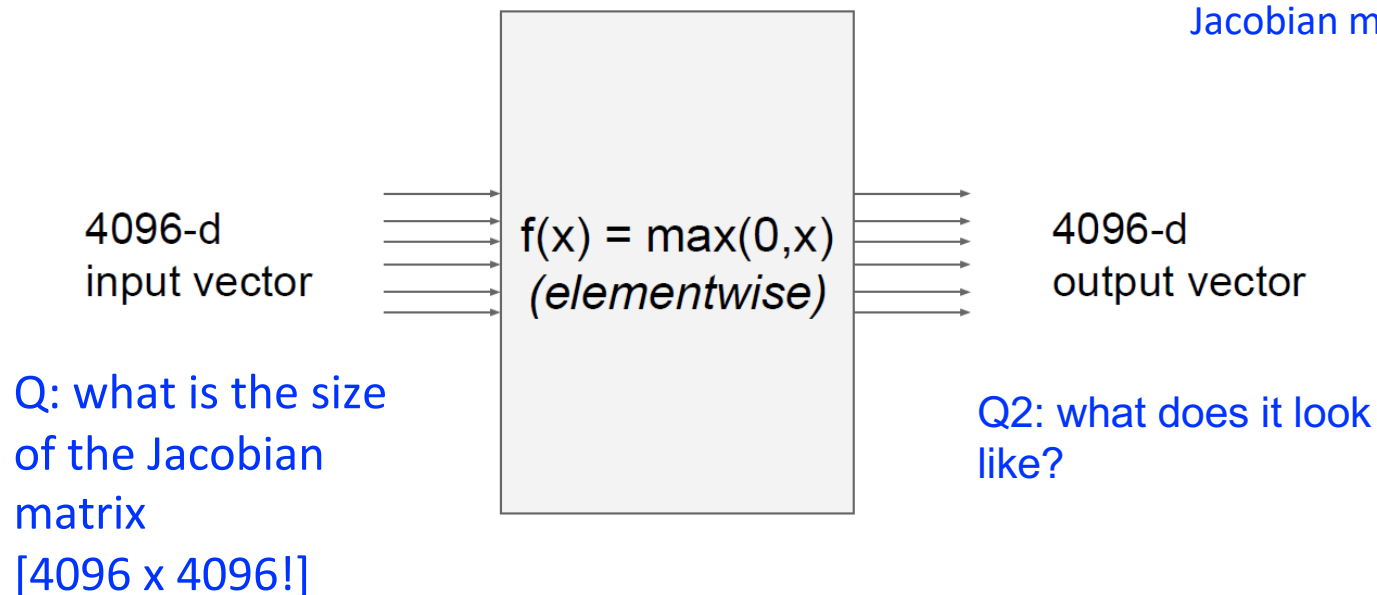


Gradients for vector

Vectorized operation

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Jacobian matrix

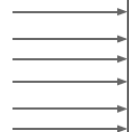


Gradients for vector

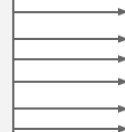
Vectorized operation

in practice we process an entire minibatch (e.g. 100) of examples at one time:

100 4096-d
input vectors



$f(x) = \max(0, x)$
(*elementwise*)



100 4096-d
output vectors

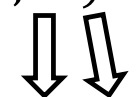
i.e. Jacobian would technically be a
[409,600 x 409,600] matrix :\\

Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

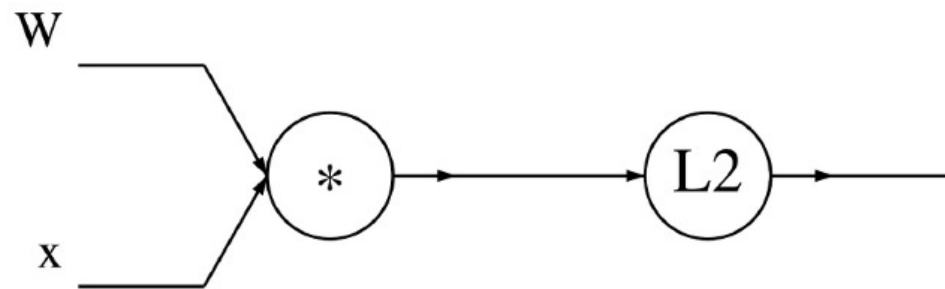
Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$


 $\in \mathbb{R}^n \quad \in \mathbb{R}^{n \times n}$

Gradients for vector

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Gradients for vector

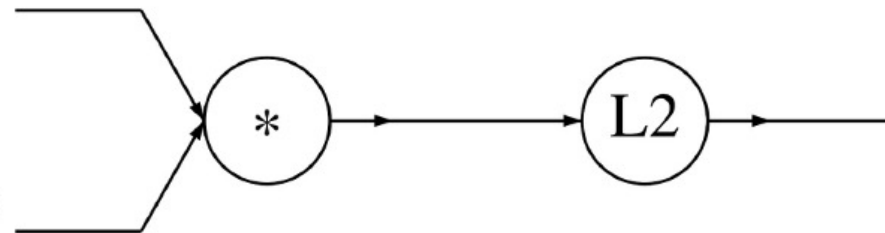
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x

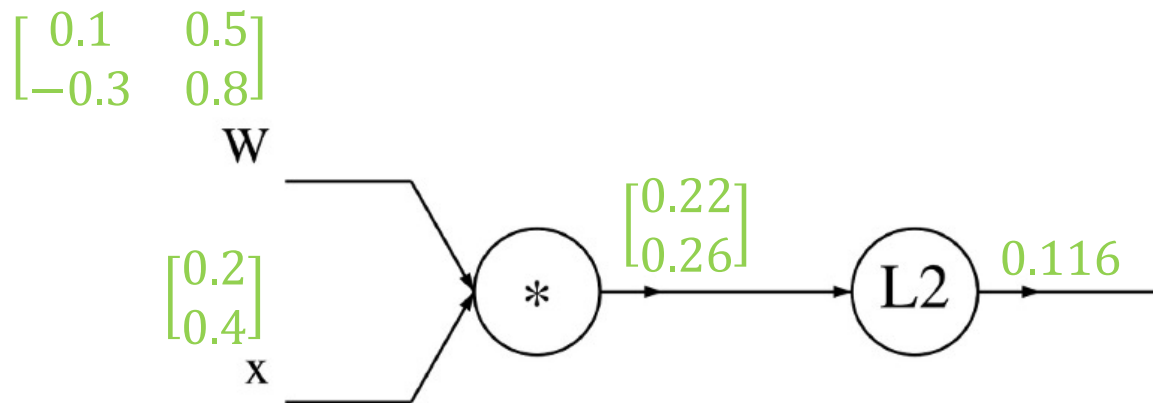


$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \cdots + q_n^2$$

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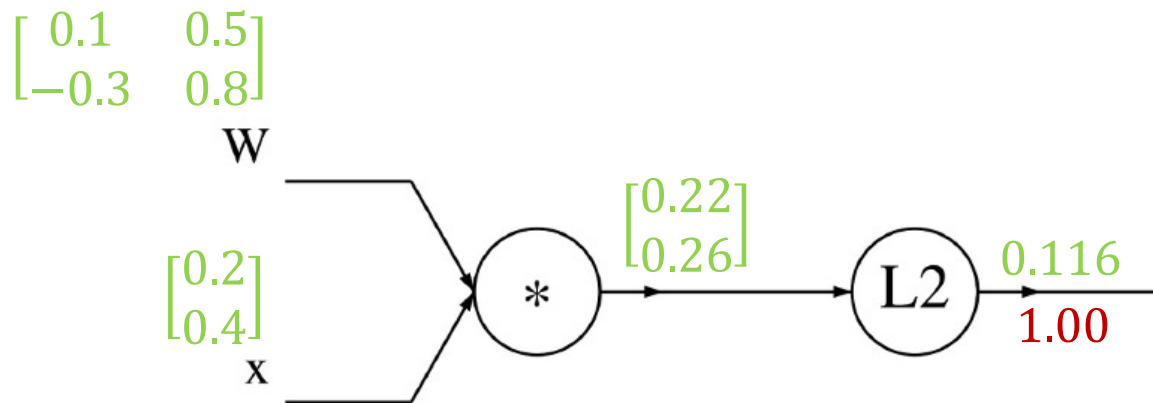


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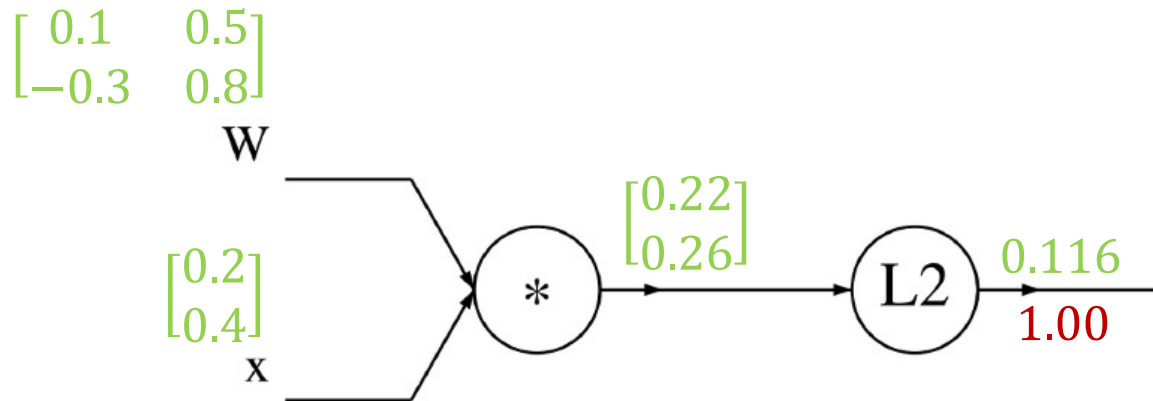


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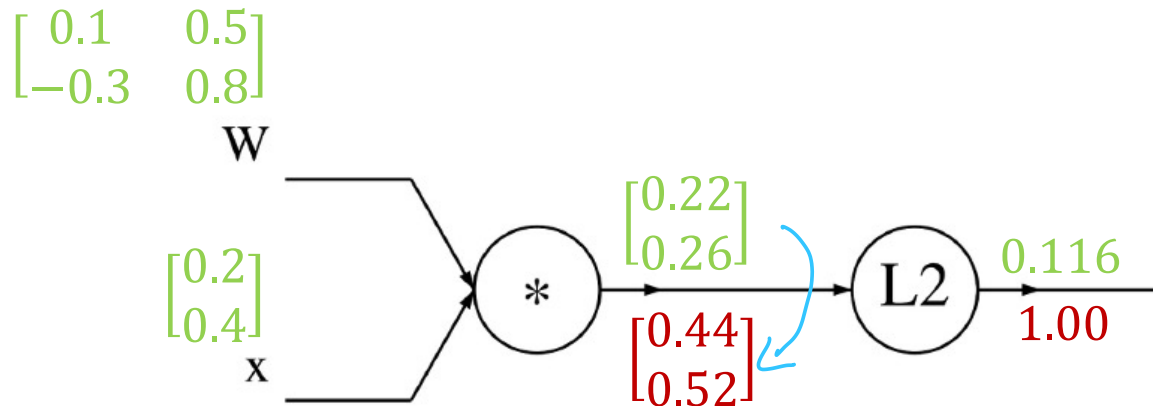
$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\frac{\partial f}{\partial q_i} = 2q_i$$

$$\boxed{\nabla_q f = 2q}$$

Gradients for vector

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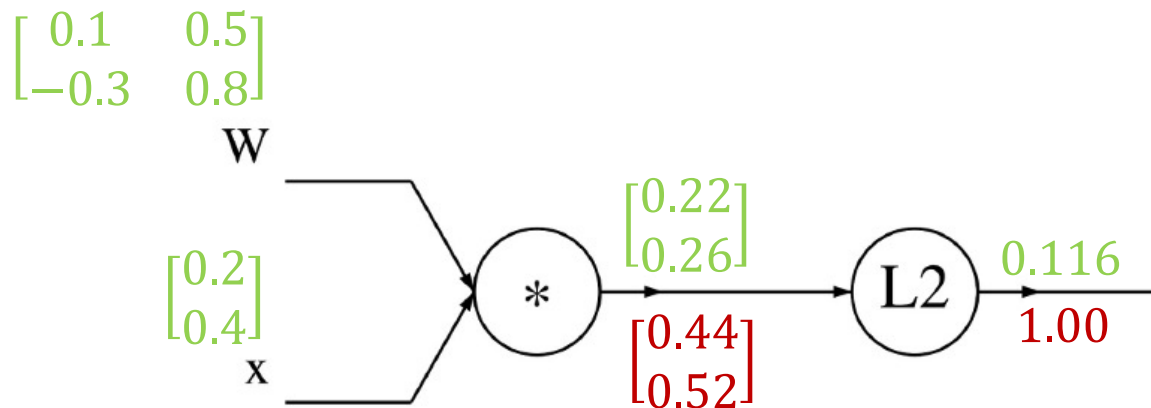
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$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

$$\frac{\partial q_i}{\partial W_{ij}} = \frac{W_{i,1}x_1 + \dots + W_{i,n}x_n}{\partial W_{ij}} = x_j$$

Gradients for vector

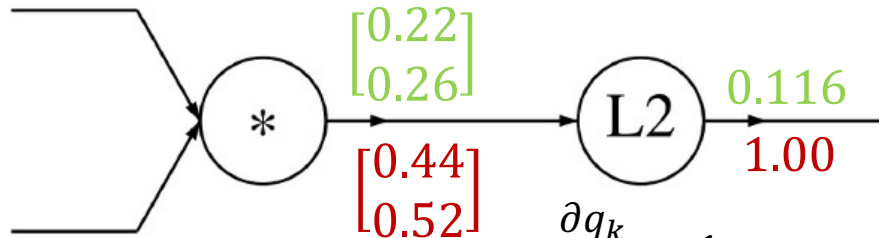
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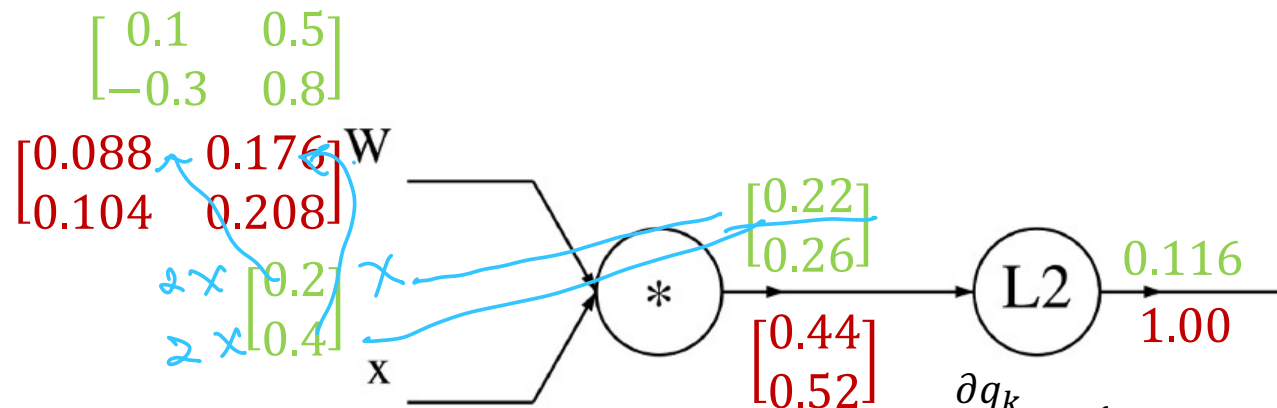
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$$\begin{aligned} \frac{\partial q_k}{\partial W_{i,j}} &= \mathbf{1}_{k=i} x_j \\ \frac{\partial f}{\partial W_{i,j}} &= \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}} \\ &= \sum_k (2q_k) (\mathbf{1}_{k=i} x_j) \\ &= 2q_i x_j \end{aligned}$$

Gradients for vector

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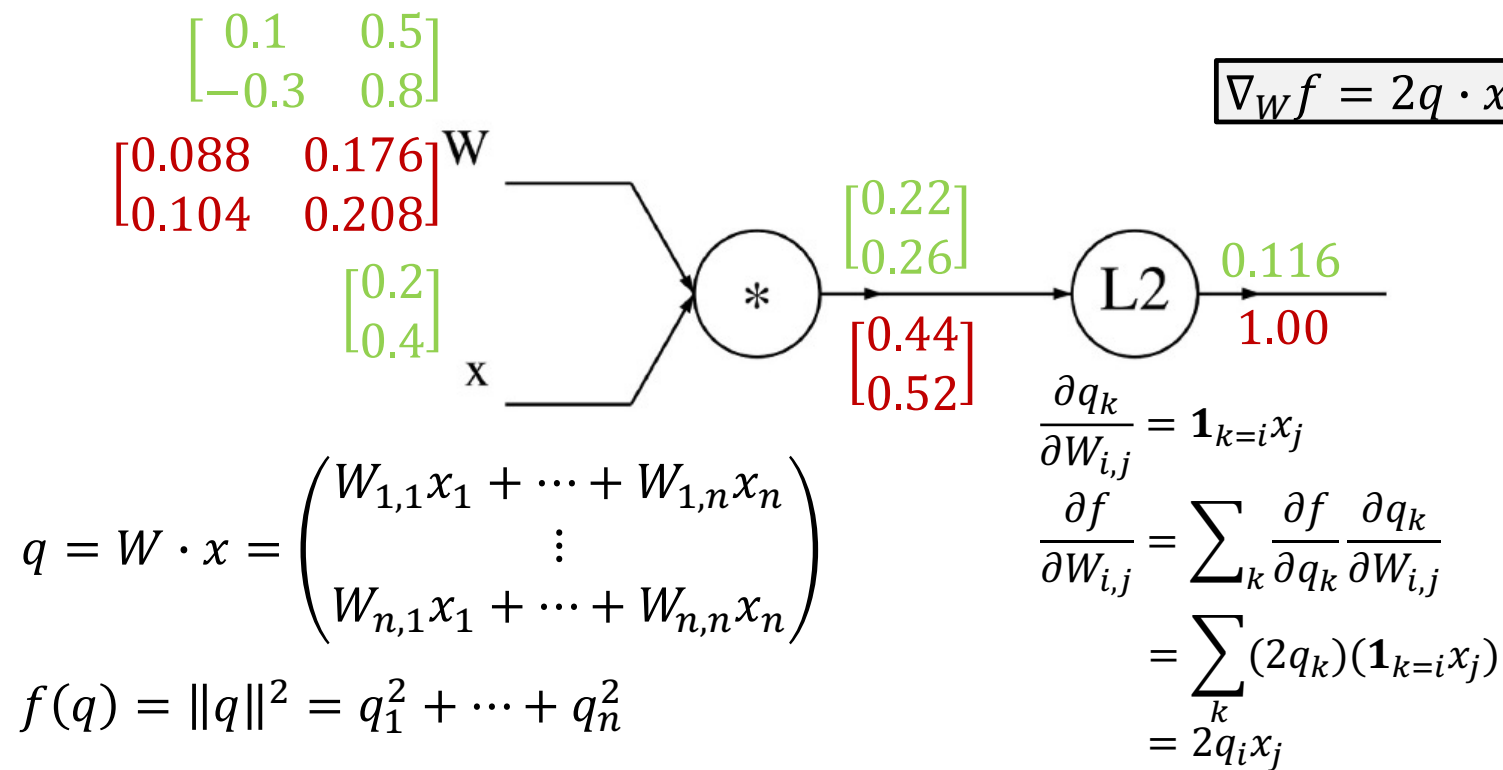
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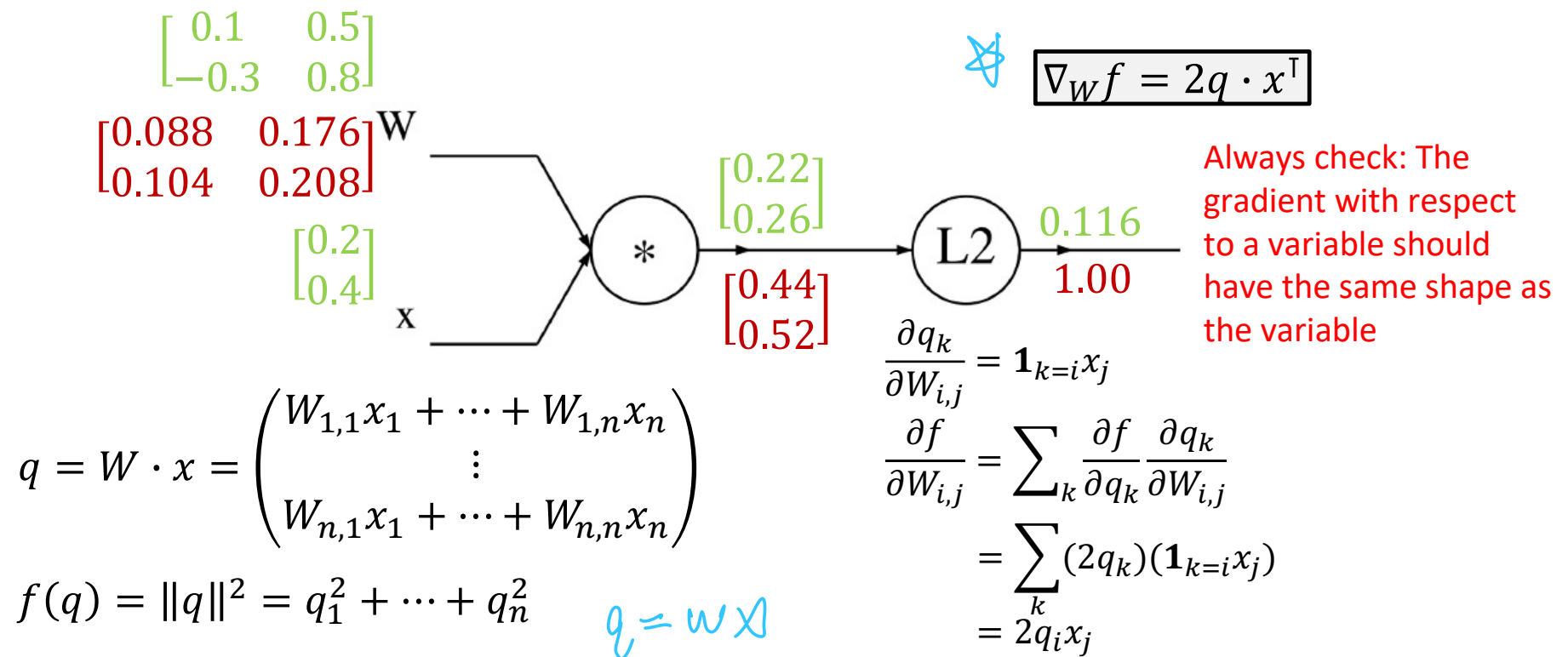
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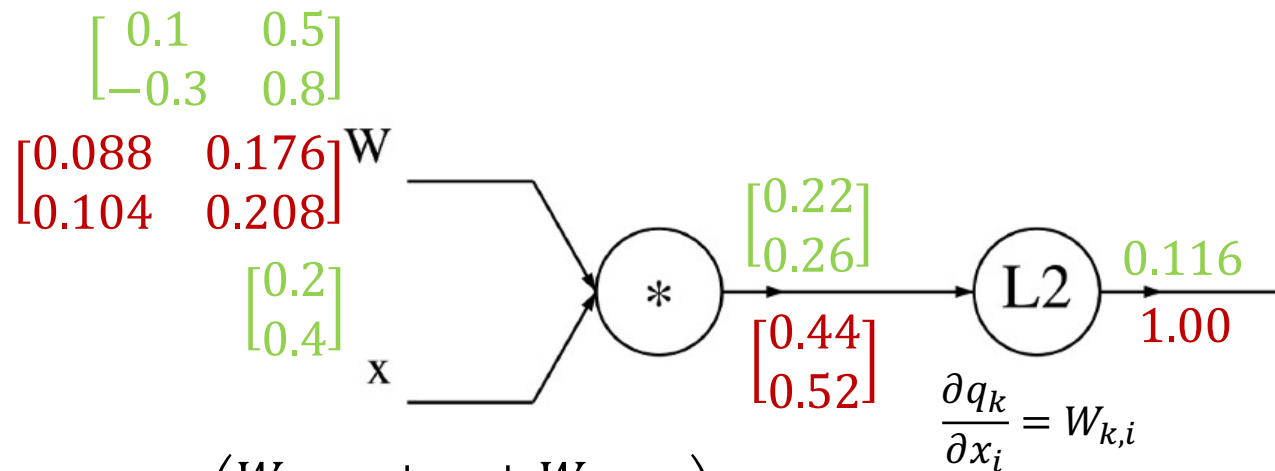
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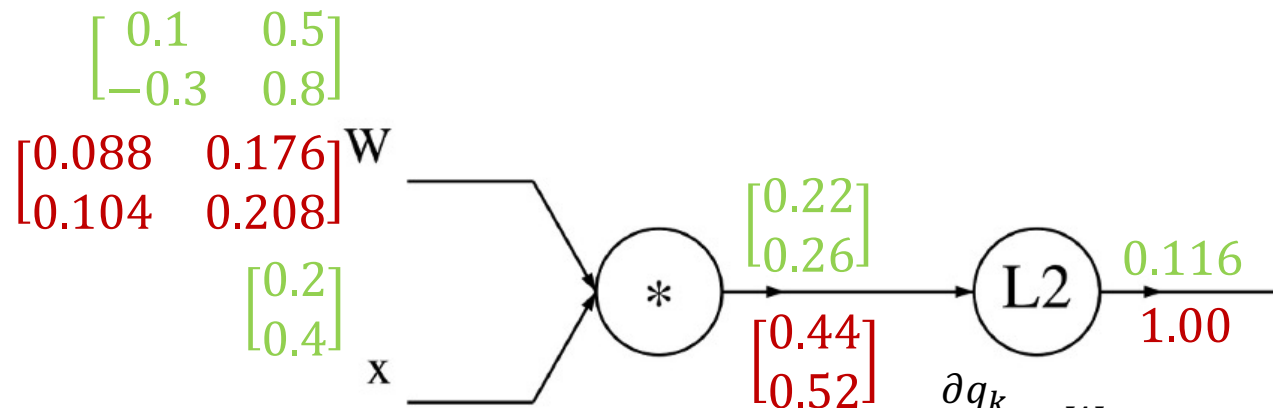


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Gradients for vector

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